

## Project Design Phase-II

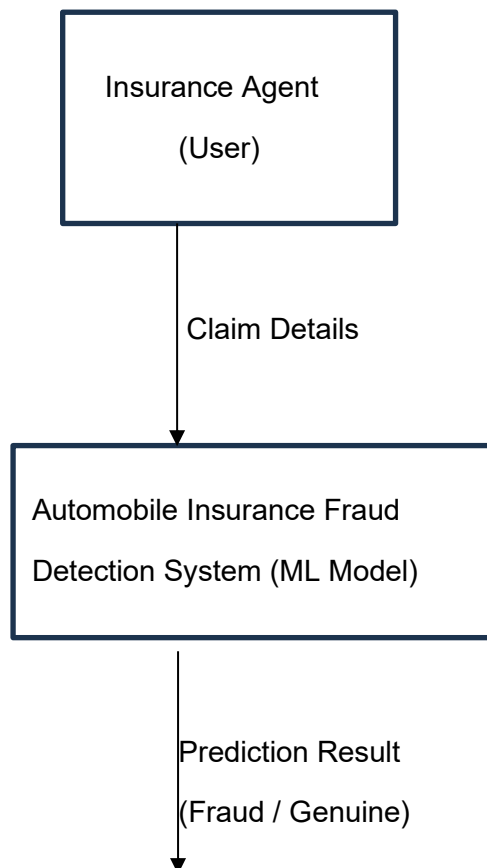
### Data Flow Diagram & User Stories

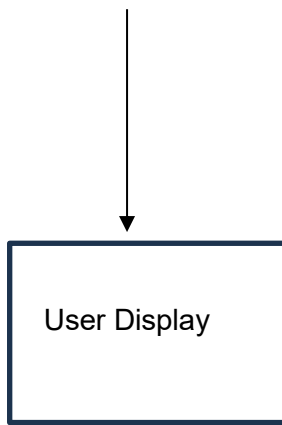
|               |                                             |
|---------------|---------------------------------------------|
| Date          | 9 March 2026                                |
| Team ID       | SWUID202641986                              |
| Project Name  | Automobile Insurance Fraud Detection System |
| Maximum Marks | 4 Marks                                     |

#### Data Flow Diagrams:

##### Level 0 DFD (Context Diagram)

This diagram shows the **overall system interaction with the user**.



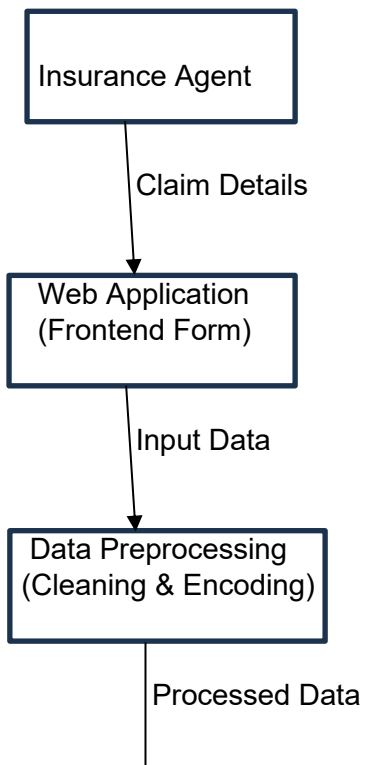


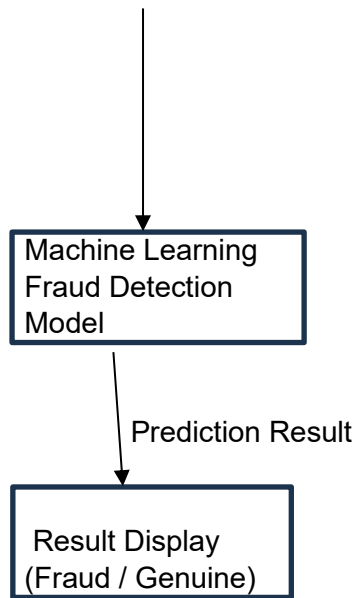
### Explanation

- The **Insurance Agent** enters automobile claim details.
- The **Fraud Detection System** processes the information using a trained machine learning model.
- The system returns the **prediction result (Fraud or Genuine)**.

### Level 1 DFD

This diagram shows the **internal working process of the system**.





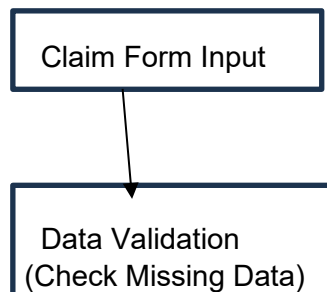
### Explanation

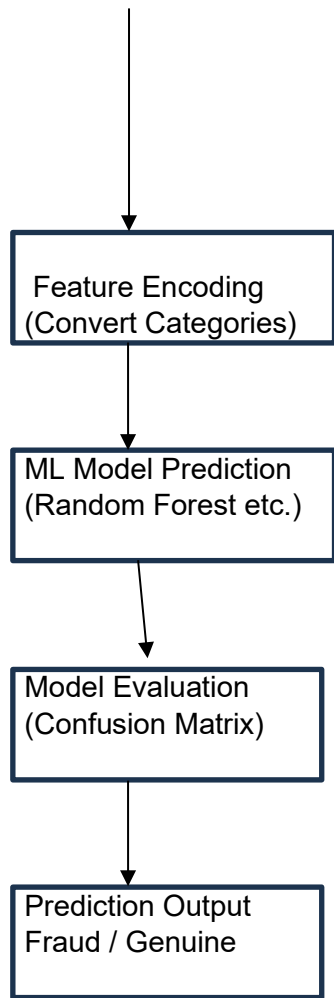
1. The **Insurance Agent** enters claim information in the form.
2. The **Web Application** collects the data.
3. **Data Preprocessing** prepares the data (cleaning, encoding).
4. The **Machine Learning Model** analyzes the data and predicts fraud.
5. The **Result Display** shows whether the claim is **Fraudulent or Genuine**.

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### 3. Level 2 DFD (Detailed Process)

This diagram shows a **more detailed flow of the fraud detection process**.





### Explanation

- **Data Validation** ensures correct input.
- **Feature Encoding** converts categorical values into numeric format.
- The **Machine Learning Model** predicts fraud.
- **Confusion Matrix** helps evaluate model accuracy.
- The system provides the **final prediction result**.

## Examples:

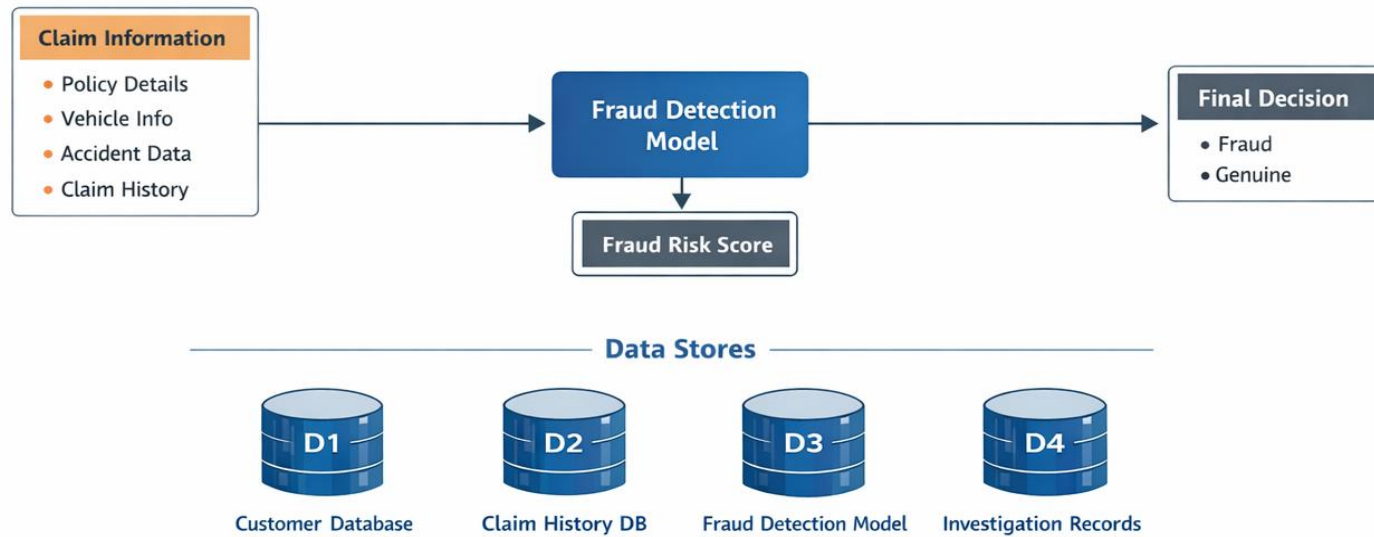
### Level 0: Context Diagram



### Level 1: Process Overview



### Level 2: Detailed Processing



## User Stories

Use the below template to list all the user stories for the product.

| User Type              | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                          | Acceptance criteria                                                   | Priority | Release  |
|------------------------|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------|----------|
| Customer (Mobile user) | Registration                  | USN-1             | As a user, I can register for the application by entering my name, email, password, and phone number.                      | User account is created and confirmation message is shown.            | High     | Sprint-1 |
| Customer (Mobile user) | Registration                  | USN-2             | As a user, I receive a confirmation email after registration.                                                              | User can click confirmation link and activate account.                | High     | Sprint-1 |
| Customer (Mobile user) | Login                         | USN-3             | As a user, I can log into the system using email and password.                                                             | User is redirected to the dashboard after successful login.           | Low      | Sprint-1 |
| Customer (Mobile user) | Claim Status                  | USN-4             | As a user, I can submit automobile insurance claim details such as vehicle information, accident details, and claim amount | Claim details are successfully stored in the system.                  | Medium   | Sprint-2 |
| Customer (Mobile user) | Claim Submission              | USN-5             | As a user, I can check the status of my submitted claim.                                                                   | System displays claim status (Pending / Approved / Fraud Suspicious). | High     | Sprint-2 |
| Insurance Agent        | Claim Verification            | USN-6             | As an insurance agent, I can review submitted claim details.                                                               | Agent can view claim information and supporting documents.            | High     | Sprint-2 |
| Insurance Agent        | Fraud Analysis                | USN-7             | As an agent, I can run fraud detection analysis on submitted claims.                                                       | System generates fraud probability or fraud risk score.               | High     | Sprint-3 |
| Insurance Agent        | Claim Decision                | USN-8             | As an agent, I can approve or reject claims based on system results.                                                       | Claim status updates and notification is sent to the user.            | High     | Sprint-3 |
| Administrator          | Data                          | USN-9             | As an admin, I can manage user accounts and claim records.                                                                 | Admin can add, edit, or delete records.                               | Medium   | Sprint-3 |
| Administrator          | Model Monitoring              | USN-10            | As an admin, I can monitor fraud detection reports and system performance.                                                 | System shows analytics and fraud detection statistics.                | Medium   | Sprint-4 |
|                        |                               |                   |                                                                                                                            |                                                                       |          |          |
|                        |                               |                   |                                                                                                                            |                                                                       |          |          |